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SIMATS ENGINEERING

ASSESSMENT TOOL - 3 : Class test

COURSE NAME : Computer Vision with OpenCV

COURSE CODE : DS-A02

COURSE OUTCOME COVERED

CO2: Apply image enhancement & segmentation techniques for extracting meaningful regions from images.

Assessment Tool weightage : 10%

(1) An image appears too bright. Explain & justify the most suitable grey level transformation to improve visibility.

(2) A dark image lacks details. Describe & justify an appropriate grey level transformation to enhance visibility

(3) An image has poor contrast. explain a suitable histogram based technique for contrast enhancement.

(4) A low-constrast image requires uniform intensity distribution. Describe & justify the method you would apply.

(5) An image shows uneven brightness across regions. explain & apply an appropriate enhancement technique.

(6) A noisy image contains random intensity variations. Explain & justify the most suitable smoothing filter.

7. An image require noise reduction without significant loss of details. Describe & apply an appropriate filter approach.

8. An image appears blurred after smoothing. Explain & justify a suitable sharpening method to restore details.

9. fine details in an image are not clearly visible.

10. A noisy image must be preprocessed before further analysis. Explain & apply a suitable spatial filter method.

Code of Conduct

I A. Pavithra certify that this submission is my original work & that I have adhered to a guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

1A) (a) Interpretation

An image appears too bright because most pixel intensity value are close to the maximum graylevel.

(b) Suitable Technique

Power-law Transformation with $\gamma > 1$

(c) Concept

Power-law transformation modified the brightness because most pixels intensity value are close to the maximum graylevel.

(d) Working Principle

(1) Read the input image.

(2) Apply the gamma transformation to every pixel.

2A) A dark image contains mostly low-intensity pixel values important objects become diffecult to recognize because shadow regions domains the image

(a) Interpretation

A dark image contains mostly low intensity pixel values important objects become difficult to recognize because shadow

(b) Suitable Technique

Log Information. The dark image becomes brighter

③ ⓐ Interpretation.

The image has a narrow intensity range causing poor constrast between objects background.

ⓑ suitable Technique.

Histogram equalization

The Image has improved contrasts making objects & edges easier to identify.

④ ⓐ Intepretation

Random variation in intensity indicates the presence of Image noise, often caused by sensors or transmission erros.

ⓑ suitable filter

4 Gaussian filter

The Image becomes smoother with reduced random noise while presering Important features.

⑤ ⓐ Intepretation

The Image contains noise, but Important edges & fine details must remain unchanged.

ⓑ Suitable filter

Median filter is nonlinear filter that replaces the center pixel with the medium value.

6A) (a) Intapretation

Excessive smoothing reduces high frequency information causing blurred edges and loss of details.

(b) Suitable Technique

duplication shaping

(c) Concept

The duplacian opeator detects rapid intensity changes using the second deivation and enhances edges.

(d) Expected Outcome.

shaper edges and improved overall image quality

(7) (a) Intapretation.

High-frequency noise produces rapid changes in pixel intensity and reduces image quality

(b) Suitable filter

low-pass filta (LPf)

(c) Concept

A low-pass filta removes high-frequency components while retaining low-frequarncy Information representing the main image content.

(1) Compute the fourier Transform

(2) Apply the low-pass files.

8) a) Interpretation

The foreground object has higher intensity than the background making segmentation easier.

b) Suitable Technique

Global Thresholding.

c) Concept

Threshold separate an image into foreground & background image using single threshold value.

9) a) Interpretation.

Edge represent boundaries where pixel intensity changes rapidly.

b) Suitable Technique.

Sobel Edge Detection

c) concept

The sobel operator calculates horizontal & vertical gradient to detect Image edges.

10) a) Interpretation

Applications such as Medical Imaging robotics & quality insepectation enquire accurate extraction of object boundaries.

b) Suitable Techniques.

Image segmentation followed by Canny Edge Detection.